Apparently Impossible Halting Problems

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Halting problems that are **surprisingly** semi-decidable

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Does program P halt ?

The Classical Case

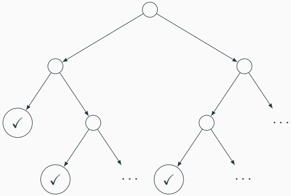
Does program P halt ?

Clearly semi-decidable

Just run the program until it halts

Non-Deterministic Computation

Does program P halt **for all** schedulers in Cantor $\{0,1\}^\omega$?



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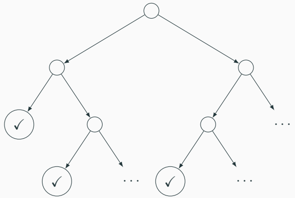

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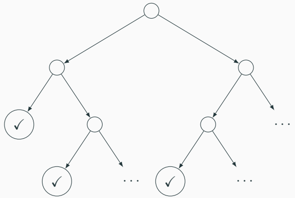

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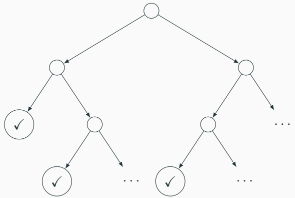

Still semi-decidable

Eureka 1 : search over $\{0,1\}^\omega$ **breadth-wise**

Eureka 2 : apply **König's Lemma**

Mixed Non-Deterministic Computation

Harbours both non-deterministic and probabilistic operations

Used e.g. in concurrent probabilistic languages

Halting $\Longrightarrow$ Statistical termination

Mixed Non-Deterministic Computation

Does P halt **for all** schedulers in Hilbert $[0,1]^\omega$ with prob $> p$?

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related to Annabelle and Carroll's work in the 00's

Eureka 1 : perform a certain **minimisation** procedure

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Eureka 1 : perform a certain **minimisation** procedure

Eureka 2 : apply a **topological** generalisation of König's Lemma

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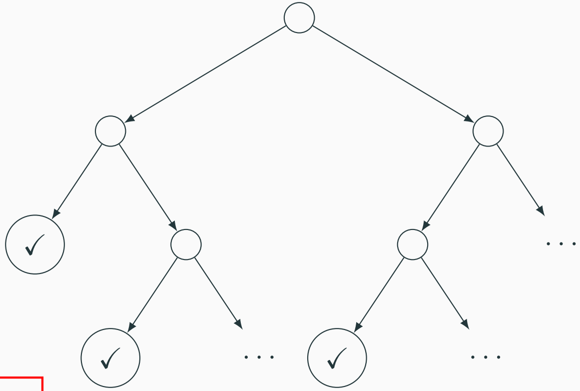
König's Lemma



i.e. P halts for every sch.

If every path is finite and the tree is finitely branching then it has finite height

$$height(root) = 1 + \bigvee_{\{s \mid root \to s\}} height(s) \tag{1}$$

Proof by contradiction

Suppose $height(root) = \infty$

Since (1) finitely branching ensures at least one son with infinite weight

Repeat this observation along the tree to build an infinite path

The Problem of Mixed Non-Determinism

Whilst previously at every step the number of choices was finite

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The Problem of Mixed Non-Determinism

Whilst previously at every step the number of choices was finite

... now it is **uncountable**

... but still **compact**

recall Hilbert cube $[0, 1]^\omega$

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Basic Ingredients

A Hausdorff topological space X

An open $\phi \subseteq X$ (an **observation**)

A continuous map $X \to \mathrm{P}_c(X)$ (the **small-step operational semantics** of a lng)

Hyperspace of compact, non-empty subsets

Our Goal

Topological König's Lemma

Assume that for every path we observe ϕ in a finite number of steps

Then there is a finite upper bound on

$$\left\{ n \in \mathbb{N} \mid p \in X^{\omega} \text{ is a path that reaches } \phi \text{ in } n \text{ steps} \right\}$$

Thought Experiment: Consequences of Paths Compactness

$$\begin{aligned}
\mathrm{Paths} &\subseteq \bigcup_{i \in \mathbb{N}_+} X^{i-1} \times \phi \times X^{\omega} && \{\text{All paths reach } \phi\} \\
&\subseteq \bigcup_{i \in F} X^{i-1} \times \phi \times X^{\omega} && \{\text{Compactness}\}
\end{aligned}$$

$\max F$ is the aforementioned upper bound !!

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$\max F$ is the aforementioned upper bound !!

Thus the next slides will be devoted to showing compactness

The Quest for Compactness

Step 1

We regard $\mathrm{Paths} \subseteq X^\omega$ as a limit of

$$\mathrm{Paths}(1) \xleftarrow{\mathrm{lst}} \mathrm{Paths}(2) \xleftarrow{\mathrm{lst}} \ldots \qquad (\mathrm{lst}(x_1 \ldots x_{i+1}) = x_1 \ldots x_i)$$

in the category of compact Hausdorff spaces

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in the category of compact Hausdorff spaces

Ofc we still need to show that every $\mathrm{Paths}(i)$ is compact Hausdorff

Step 2: The continuous ext function

$$
\begin{aligned}
X^i &\cong X^{i-1} \times X \\
&\xrightarrow{\mathrm{id}\times\Delta} X^{i-1} \times X \times X \\
&\xrightarrow{\mathrm{id}\times\mathrm{op}} X^{i-1} \times X \times \mathrm{P}_c(X) \\
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$\mathrm{Paths(i)}$ compact $\Longrightarrow \mathrm{ext}^\star(\mathrm{Paths(i)})$ compact $\Longrightarrow \mathrm{Paths(i+1)}$ compact

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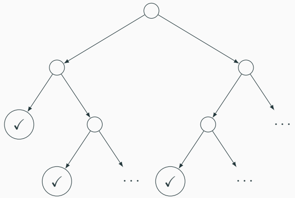

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X a discrete topological space

S a discrete topological space of **halting states**

Observations ϕ defined as $\emptyset + U \subseteq X + S$ where $U \subseteq S$

Instantiating the Framework

X a discrete topological space

S a discrete topological space of **halting states**

Observations ϕ defined as $\emptyset + U \subseteq X + S$ where $U \subseteq S$

resumptions model

finite, non-empty subsets

$$\frac{\mathrm{op} : X \to \mathrm{P}_c(X + S)}{[\mathrm{op}, \eta \cdot \mathrm{inr}] : X + S \to \mathrm{P}_c(X + S)}$$

lifting to the algebras of $(-) + S$

A Sketch of the Semi-Decision Procedure

Exhaustive Search

Regarding $\mathrm{Paths} \subseteq X^\omega$ as a limit of

$$\mathrm{Paths}(1) \xleftarrow{\mathrm{lst}} \mathrm{Paths}(2) \xleftarrow{\mathrm{lst}} \dots$$

we check every limit component until finding one in which all paths reach ϕ

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We will need a series of domain-theoretic notions

Take a **domain** X and let $\mathcal{O}(X)$ be its **Scott** topology

Definition (Continuous valuation)

A map $\mu : \mathcal{O}(X) \to [0, \infty]$ that satisfies certain conditions

Probabilistic Powerdomain

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Definition (Continuous valuation)

A map $\mu : \mathcal{O}(X) \to [0, \infty]$ that satisfies certain conditions

The space of continuous valuations $\mathcal{V}(X)$ is equipped with a topology generated by the opens (**statistical observations**)

$$\odot_p U \triangleq \{\mu \mid \mu(U) > p\} \qquad (U \text{ an open of } X, \text{ and } p \in [0, \infty[)$$

Objects are domains with the operations of a **cone**
(plus some compatibility conditions)

Morphisms are **Scott**-continuous **linear** maps

The Category of Algebras of $\mathcal{V}(-)$

Objects are domains with the operations of a **cone**
(plus some compatibility conditions)

Morphisms are **Scott**-continuous **linear** maps

It has all products

It features a monad of compact subsets

The monad’s introduction requires a few preliminaries

Geometric Convexity

Definition (Geometric convex closure)

Subset A of a cone C is called geometrically convex closed if

$$a, b \in A \Longrightarrow p \cdot a + (1 - p) \cdot b$$

Geometric convex closure of a subset $A \subseteq C$ is defined by

$$conv\, A = \left\{ \sum_i p_i \cdot a_i \mid \sum_i p_i = 1 \text{ and } \forall i \in I.\, a_i \in A \right\}$$

Subsets in the Geometrically Convex Powercone $\mathrm{P}_b(-)$

These must have the following properties

- non-empty
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- Lawson-compact
- ...

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It suffices to work with just those subsets of the form

$$\overline{conv\,F} \cap \uparrow conv\,F \qquad (F \subseteq \mathcal{V}(S) \text{ a finite set of valuations})$$

Theorem

Let X be a zero-dimensional domain. Then

$$\mathit{conv}\, F = \overline{\mathit{conv}\, F} \cap \uparrow \mathit{conv}\, F \qquad (F \subseteq \mathcal{V}(X)_{\leq 1} \text{ a finite set of valuations })$$

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Example

If $F = \{\mu_1, \mu_2\}$ then $conv\, F = \left\{p \cdot \mu_1 + (1 - p) \cdot \mu_2 \mid p \in [0, 1]\right\}$

the compact branching we saw with sch. in Hilbert $[0, 1]^\omega$

Instantiating the Framework

X a discrete domain

S a discrete domain of **halting states**

Statistical observations defined as $\odot_p(\emptyset + U)$ where $U \subseteq S$

$$\cfrac{\cfrac{\mathrm{op} : X \to \mathrm{P}_b\mathcal{V}(X+S)}{[\mathrm{op}, \eta \cdot \mathrm{inr}] : X + S \to \mathrm{P}_b\mathcal{V}(X+S)}}{[\mathrm{op}, \eta \cdot \mathrm{inr}]^\star : \mathcal{V}(X+S) \to \mathrm{P}_b\mathcal{V}(X+S)}$$

lifting to the algebras of $\mathcal{V}(-)$

Hausdorff in the Lawson topology

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However ...

The Minimisation Procedure

Theorem

$$\mathit{conv}\, F \subseteq \odot_p U \iff \bigwedge_{\mu \in F} \mu(U) > p$$

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Conclusions

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Can we obtain an efficient semi-decision procedure from these results ?

Can they be generalised to the level of a topos ?

Renato Neves.
An adequacy theorem between mixed powerdomains and probabilistic concurrency.
In Cinzia Di Giusto and Giorgio Bacci, editors, *Proceedings of the Combined 32nd International Workshop on Expressiveness in Concurrency and 22nd Workshop on Structural Operational Semantics, EXPRESS/SOS 2025, and 22nd Workshop on Structural Operational SemanticsAarhus, Denmark, 25th August 2025*, EPTCS, pages 23–45, October 2025.